The Arrow's Engine: The Dynamical Rule of Event-Density Gravity, and the Closing of Its Weak-Field Assumptions

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Abstract

This paper closes the two weak-field assumptions of GR-I, builds the dynamical rule GR-II reduced the field equation to, and resolves KM-II §6's open scalar-mode question. GR-I derived Event Density's weak-field Einstein metric under two labeled assumptions — the commitment-rate linearity `alpha = 1` (selecting the Einstein over the Nordström branch) and the timelike geodesic identity (that massive matter free-falls on the bandwidth metric). GR-II established that ED's gravity is khronometric and reduced the field equation, by Lovelock's theorem, to *the steady state of an admissible dynamical-bandwidth rule* `b-dot = F`, leaving `F` unbuilt and the khronon's scalar-mode speed open (KM-II §6). This paper builds `F`, closes both assumptions, and fixes the khronon speed — with one spine running through all of it: **the arrow (P11 irreversibility) selects the Einstein/khronometric structure three independent times.** (i) **`alpha = 1` is forced** from the P04 band law: the commitment-reserve is a distinct, *monotone-draining* band, and the Nordström branch would require it to replenish in step with the metric band — which P11 forbids; Nordström is positively excluded (modulo a named band-accounting premise). (ii) **The timelike geodesic identity is reduced** to the massive eikonal of GR-I's proven null result — the massive Jacobi functional is limit-forced to recover both the null (Fermat) and Newtonian (Maupertuis) limits, with the explicit Σ-rule computation the residual. (iii) **The khronon propagates at the speed of light** (`c_s = c`): the only candidate for a foliation-specific kinetic term that would give it a second cone is the reserve sector, and P11 makes that sector dissipative — it *damps* the khronon (confirmed in simulation: it damps and overdamps without shifting the cone), it does not supply the conservative `lambdatheta^2` term a second cone needs; this resolves KM-II §6 toward maximal predictivity. The rule is then **built and run**: in its forced form (`b-dot = Dgrad ^2b - kapparho`, P02 adjacency-sharing minus P11 matter-concentration sink) its steady state is the Newtonian field equation `grad ^2b ~ rho` (the fixed point of the two forced terms); the deficit amplitude is exactly linear in source (Schwarzschild `r_s ~ M`, confirmed in 2D and 3D); strong coupling forms a **finite-radius `b -> 0` horizon** that is a **frozen A2 decoupling cut** — realizing dynamically the identity that the A2 cut, the metric horizon, and the `b -> 0` locus are one object; the emergent horizon carries **area-law (holographic) entropy `S ~ A`**; and the hyperbolic rule carries **2 tensor + 1 scalar** modes (the mode count read off `F` rather than the gauge group). The emergent horizon's thermodynamics follow: area-law `S ~ A` (measured) and the Hawking scaling `kappa = 1/(2r_s) ~ 1/r_h` (derived from the vacuum solution `b = 1 - r_s/r` + GR-I's `g_00g_rr = -1`, with `r_s ~ M` measured). One number remains honestly open and is flagged in-paper: the preferred-frame

parameters `alpha_1, alpha_2` (ED is driven into the maximally-suppressed corner by five independent mechanisms, but the *values* are not computed and not faked). The result: **the rule ED's gravity reduces to is built, its weak-field assumptions are closed, and its scalar mode is pinned to the light cone — with the same irreversibility doing all three jobs.** Einstein is not derived; one falsification number (`alpha_1, alpha_2`) remains open; the engine exists and runs.

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1. Introduction

1.1 What this paper does

GR-I and GR-II took ED's gravity to the khronometric class with the weak-field Einstein tests — but on two labeled assumptions and with the central object, the dynamical rule, unbuilt. This paper closes that. It (a) **forces `alpha = 1`** from the P04 band law (§4), (b) **reduces the timelike geodesic identity** to a proven limit (§5), (c) **derives the khronon speed `c_s = c`** (§6, resolving KM-II §6), and (d) **builds and runs the dynamical rule `F`** (§7), confirming the Newtonian field equation, the Schwarzschild mass-scaling, an emergent frozen horizon, the area law, the Hawking scaling, and the mode count. It then states honestly the **one number that remains open** (§8): the preferred-frame `alpha_1, alpha_2` (the Hawking scaling, earlier thought open, is corrected there to a derived result).

1.2 Why this matters, and the register

The register of this paper is *closing the gaps*, not breakthrough. Its value is that the quartet's two standing weak-field assumptions become derived (or reduced), the object the whole construction reduces to is shown to exist and run, and a published open question (KM-II §6) is answered. The conceptual spine is worth stating plainly: **one primitive — the arrow, P11 irreversibility — selects the Einstein/khronometric structure three separate times** (the lapse, the mode count, the scalar speed). That the factor of two, the extra scalar, and that scalar's speed all trace to one commitment is the paper's organizing fact.

1.3 How this fits the arc

- **GR-I:** the weak-field metric and classical tests, on the `alpha=1` and geodesic assumptions. *(this paper closes both)*
- **GR-II:** the khronometric class; the field equation reduced to the steady state of `F` (unbuilt); the scalar speed open. *(this paper builds `F`; fixes the speed)*
- **KM-I/KM-II:** the deep-field (MOND) and cosmological sectors, inheriting the weak-field metric. *(KM-II §6 resolved here)*
- **GR-III (this paper):** the engine — the dynamical rule built, the weak-field assumptions closed, the khronon speed pinned, one number (`alpha_1, alpha_2`) left open.

1.4 Conventions and regime of validity

Signature `(-,+,+,+)`; `c = 1` where convenient. The derivations are **weak-field / DCGT-limit**; the simulations are minimal-model (modest grids; coefficients `kappa, D` unpinned). Where a result is *measured* it is labeled so; where *forced/derived/reduced* it is tiered; where *open* it is flagged. No strong-field claim is made beyond the emergent-horizon structure; the `alpha_1, alpha_2` values are explicitly deferred, and the Hawking scaling is derived (the rule's vacuum solution + GR-I) with the direct sim measurement resolution-limited.

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2. Primitive Inputs

Substrate primitives (postulated, Paper_087), load-bearing: P11 (commitment irreversibility — the arrow; load-bearing three times), P13 (the tick), P04 (the four-band bandwidth, including the commitment-reserve band), P02 (reciprocal adjacency — the shared metric band), P05 (transport — the single causal cone).

Inherited results: GR-I (`g ~ b^-1`, `N^2 ~ b`, the weak-field metric; the `alpha=1` and geodesic assumptions this paper closes); GR-II (the khronometric class; the field equation = steady state of `F`; the mode-count gauge argument); Paper_027 (`G`, value-inherited); R2 (the dynamical rule's admissibility); R4–R9 (the structural arc: genuine metric, conserved source, Lovelock).

External dictionary (inherited, labeled I): Lovelock's theorem (GR-II); the khronometric/Einstein-aether mode and PPN structure; the 3D-vs-2D CFL stability bound for explicit diffusion.

No new primitive is introduced; no primitive forcing is invoked. The dynamical rule is the formalization of the commitment-reserve dynamics P04 + P11 already declare (R2).

2.5 Load-Bearing Step Audit

Status legend: **I** = inherited; **D** = derived here; **D-cond** = derived modulo a named premise/limit; **measured** = simulation result; **structural** = from structure alone; **fixed-point** = the steady state of the forced terms (built-in via the terms being primitive-grounded); **open** = declared open, not computed.

Step	Status	Source
field equation = steady state of `b-dot = F`	**I**	GR-II R9
`alpha = 1` forced (Nordström excluded)	**D-cond** (band-accounting premise)	§4
Nordström needs reserve replenishment → inadmissible (P11)	**structural**	§4
timelike geodesic identity = massive eikonal of the null result	**D-cond** (limit-forced; Σ-computation residual)	§5
`c_s = c` (`eps = 0`, khronon at light speed)	**D** (leading order)	§6
reserve dissipative → damps, not a `lambdatheta^2` second cone	**structural + measured**	§6

Step	Status	Source
the arrow (P11) selects the structure three times	**structural unification**	§4–§6
`F` built; steady state = `grad ^2b ~ rho`	**fixed-point** (corr 0.999)	§7
`r_s ~ M` (Schwarzschild mass-scaling), 2D + 3D	**measured**	§7
finite-radius `b->0` horizon = frozen A2 cut	**measured (emergent)**	§7
`S ~ A` (area-law / holographic) on the emergent horizon	**measured**	§7
2 tensor + 1 scalar, from `F`	**D**	§7
`alpha_1, alpha_2` values	**open** — structural suppression only, not computed	§8
Hawking `T ~ 1/r_h` (`kappa = 1/(2r_s)`)	**derived** (vacuum solution + GR-I; `r_s~M` measured); direct sim resolution-limited	§8
full EFE / strong field / pure GR	**NOT CLAIMED**	preamble 2

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3. The Rule the Field Equation Reduces To

GR-II's Lovelock argument (R9) established that ED's field equation is **the steady state of an admissible dynamical-bandwidth rule** `b-dot = F(rho, grad rho, b)` — the per-edge bandwidth evolution. R2 showed this rule is admissible: it is the formalization of the commitment-reserve dynamics P04 + P11 already declare, violating no certified invariant, in its commitment-driven, monotone-conservative, orientation-blind form. The structural arc then *forced its form*: the band↔edge map (R5, by flux↔Levi-Civita covariance), the lapse exponent (`alpha=1`, §4 below), and the geodesic structure (§5). What remained was to **build it, run it, and read off its modes** — and to fix the one open coefficient of its propagating sector, the khronon speed.

This paper does those four things. The rule's minimal forced form is

> `b-dot = D grad ^2b - kappa rho`, `b >= 0`, `b -> 1` at the frame,

with `D grad ^2b` the **P02 adjacency-sharing** (the reciprocal metric band equilibrates by the graph Laplacian — the elliptic geometry sector) and `-kapparho` the **P11 commitment-concentration sink** (persistent matter holds bandwidth in its concentrated channel, depleting the metric band $\propto$ its density). The hyperbolic (propagating) sector replaces the relaxational `Dgrad ^2b` with the retarded single-P05-transport wave operator (§6–§7).

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4. Closing the Lapse Assumption: `alpha = 1` Forced

GR-I's lapse `N^2 ~ b` (hence the Schwarzschild relation `g_00g_rr = -1`, the Einstein branch) rested on `alpha = 1` in `Gamma_commit ~ b_int/reserve`. With `Gamma ~ b^alpha`, the null condition gives `g_00g_rr ~ -b^{2(alpha-1)}`: `alpha = 1` is Einstein, `alpha = 0` is Nordström (conformal, no light bending, experimentally excluded). Writing `reserve ~ b_int^beta`, `alpha = 1 - beta`: **the Einstein/Nordström fork reduces to a single question — whether the commitment-reserve band co-scales with the internal (metric) band.**

It does not, and P11 is why. The reserve and `b_int` are **different state variables**: `b_int` is the *ambient* adjacency capacity (shaped by `rho`, `grad ^2b ~ rho`); the reserve is a *carried, monotone-draining budget* set by commitment **history** (P11; no replenishment, R2 §5). Generically, then, `beta = 0`, `alpha = 1`. The Nordström value `beta = 1` requires the reserve to *track* `b_int` — to be **raised** where `b_int` is high — which is **replenishment**, forbidden by the same P11 irreversibility that defines commitment.

> **The P04 band law forces `alpha = 1` and positively excludes Nordström.** The Einstein branch at the lapse is selected by the arrow (P11): the reserve cannot replenish to track the metric band, so it does not co-scale, so `alpha = 1`. *(Modulo one band-accounting premise: that the metric band and the rate-numerator band are the same `b_int`; named, not proven.)* Had P04 defined the reserve as a fixed fraction of the total or supplied a replenishment band, ED would have been Nordström — experimentally excluded. It does neither. **This is the arrow's first hat.**

GR-I's labeled postulate P-Commitment-Linear is thereby a derived result.

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5. Closing the Geodesic Assumption: the Timelike Identity Reduced

`grad *T = 0` for ED's matter reduces (R6) to `rho a^nu` — so conservation holds iff the bandwidth worldlines are geodesics of `g ~ b^-1`. GR-I proved this for **null** fronts (Fermat, optical index `n_opt ~ b^-1` — the source of the factor of two); the **timelike/massive** case was assumed.

The timelike-geodesic functional for `g ~ b^-1` is the **Jacobi/Maupertuis** function `J(b) = b^-1sqrt(E^2 - m^2b)`, and it is *limit-forced* — it reduces to a *proven* result in each of the two checkable limits:

limit	`J(b)` reduces to	interpretation
`m -> 0`	`E/b = E*n_opt`	**Fermat** (the proven null result — so the timelike functional *contains* it)
weak field	`sqrt(2m(E_kin - V))`	**Maupertuis** (Newtonian orbits — recovers R1)

ED's subluminal max-Σ front is then identified as the **massive eikonal** of the same propagation GR-I treated masslessly (its Hamilton–Jacobi ray being the timelike geodesic; the worldline's intrinsic clock from P11 + P13).

The timelike geodesic identity is advanced from assumed to limit-forced reduction — forced to recover both the proven null limit and the Newtonian limit, the two checks any correct functional must pass (and it could have failed the `m -> 0` check; it did not). The residual is the explicit Σ-rule massive-front computation; the explicitly-assumed identity is replaced by a located, limit-checked reduction.

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6. The Khronon Speed: `c_s = c`, Derived (KM-II §6 Resolved)

GR-II's mode count (2 tensor + 1 scalar = khronometric) left the **scalar (khronon) speed** open (KM-II §6: "may propagate at a different speed in principle"). The hyperbolic rule reduces this to one coefficient: writing the propagating perturbation as `h_tt_ij = c^2grad ^2h_ij + eps*(trace)`, the khronon speed is `c_s/c = sqrt(1+eps)`, with `eps` a foliation-specific `lambdatheta^2` kinetic term. `eps = 0` → khronon at light speed; `eps != 0` → a second cone.

`eps` is derived to be zero. The only candidate source of a `lambdatheta^2` term is the commitment-reserve sector — and P11 makes that sector **one-way / dissipative** (monotone drain, no replenishment). A dissipative coupling enters the dispersion as the **imaginary** part of `omega` (damping), never the **real** part (speed); a conservative `lambdatheta^2` term shifts the real part. So P11 irreversibility makes the reserve **incapable** of being the kinetic term `eps != 0` requires. Simulation confirms the distinction is physical: a dissipative reserve term **damps and overdamps** the khronon without moving its cone (the speed is unshifted, the amplitude decays), in sharp contrast to the `eps`-knob, which cleanly shifts `c_s/c = sqrt(1+eps)`. The forced rule (P05 transport + P11 drain) supplies no `lambdatheta^2` term, and minimality forbids positing one.

> **`eps = 0, c_s = c` — derived: ED's scalar gravitational-wave polarization is at the speed of light**, damped near active matter (overdamped where commitment fires) and clean in vacuum (observationally consistent — the far-field khronon is at `c`, undamped). This resolves KM-II §6 *as a derivation*, toward maximal predictivity — sharper than generic khronometric gravity, which allows `c_s != c`. **This is the arrow's third hat** (the second being the mode count: the arrow breaks the time-gauge that would freeze the scalar, GR-II/R10).

A scalar gravitational-wave polarization *at* `c` is, in principle, **observationally distinguishable** from both pure General Relativity (two tensor modes, no scalar) and generic khronometric gravity (a scalar at `c_s != c`) — a sharp, specific signature of ED's foliation sector.

The three hats. One primitive, P11, selects the Einstein/khronometric structure three independent times: **`alpha = 1`** at the lapse (the reserve cannot replenish — §4), the **khronon physical** at the mode count (the arrow breaks the time-gauge — GR-II), and the **khronon at light speed** (the reserve is dissipative, not kinetic — this section). The factor of two, the extra scalar, and that scalar's speed are one irreversibility wearing three hats. *The same arrow that makes the khronon keeps it on the light cone.*

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7. The Engine, Built and Run

The rule is built and run (`evaluation/DynamicalBandwidth/`); results separated into *built-in* (the fixed point of the forced terms) and *emergent* (not built in).

7.1 The Newtonian field equation (fixed point)

The steady state satisfies `grad ^2b ~ rho` (`corr(grad ^2b, rho) = 0.999`, the bandwidth-Laplacian concentrated 52× on the source, harmonic vacuum). This confirms the two forced terms (P02 sharing + P11 sink) are Newtonian-consistent — the content being that they are primitive-grounded, not that Poisson is a surprise.

7.2 Schwarzschild mass-scaling `r_s ~ M` (emergent)

The deficit amplitude is **exactly linear** in the integrated source, in **both 2D and 3D** (3D: `r_s = 0.158, 0.316, 0.633` for `M = 504, 1008, 2016`). `r_s ~ M` — the Schwarzschild relation. The raw 3D `1/r` slope (`-1.68`) is finite-box/convergence-limited: **the mass-scaling is the invariant; the raw slope is not.**

7.3 The emergent horizon = a frozen A2 cut (the emergent result)

At strong coupling, `b -> 0` on a **finite-radius surface** (`g_rr ~ 1/b -> inf`) — a metric horizon — where the reserve is exhausted, so the cut is **frozen** (P11). Nothing in the rule mentions a horizon; one forms. This realizes dynamically the identity that the `b -> 0` locus is *at once* an **A2 decoupling cut**, a **metric horizon**, and a **V5 surface** — one rule, one locus, three identities. **A2-dynamical is hereby realized by build-and-run, not argued.**

7.4 Area-law entropy `S ~ A` (emergent, holographic)

The severed-information count across the frozen cut (A1: capacity-zero across the cut → the hidden DOF are the boundary adjacency channels) scales as the horizon **perimeter** (`r_h^{0.96}`), **not** the enclosed bulk (`r_h^{2.01}`). The entropy is a **surface** quantity — holographic — on a horizon the rule formed dynamically. *(Half-structural: A1 severance is a surface cut. Half-measured: the emergent horizon is a clean compact surface — the part that could have failed and did not. The `1/4` coefficient is value-inherited.)*

7.5 The mode count, from `F` (derived)

The hyperbolic rule's gauge structure gives **2 tensor + 1 scalar** — the scalar physical because the arrow breaks the time-gauge — reproducing GR-II's khronometric count by direct operator analysis rather than the gauge-group argument. Tensor modes at `c` (single P05 cone); scalar at `c` (§6).

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8. The Open Edge — One Number Open, One Corrected

Two quantities were carried as open in earlier drafts of this work; re-examination resolves one (the Hawking scaling — a measurement correction) and leaves one genuinely open (`alpha_1, alpha_2`). The paper states the open one as open rather than leaning on the structural pressure.

`alpha_1, alpha_2` (the preferred-frame front) — structurally suppressed, not computed. ED is driven into the maximally-suppressed corner of the preferred-frame parameter space by **five independent derived mechanisms**: universal (not differential) Lorentz violation makes `alpha_1, alpha_2` sub-leading (GR-II); `c_T = c` and `c_s = c` put both gravitational cones at light speed (the most-suppressed khronometric regime); the metric-only matter coupling removes the direct khronon–matter `alpha_1` source (KM-I); and — ED-specific — the **dissipative near-field** (the overdamped khronon around matter, §6) cannot build the frame-dependent near-field response `alpha_1, alpha_2` measure (P11, again). Five independent mechanisms point toward the same suppressed corner: strong pressure toward PPN-safety. **But the *values* are not computed here, and not faked** — they require the pinned couplings (`kappa, D`) plus the khronometric PPN formulas, or a direct PPN expansion of `F`. The falsification front stays **formally open as a number**; what has changed is that the computation is now definite and foliation-knob-free.

The Hawking temperature scaling `T ~ 1/r_h` — follows (derived; direct sim resolution-limited). The emergent horizon carries both the area law (`S ~ A`, §7) and the Hawking scaling. The surface gravity, given GR-I's Schwarzschild relation `g_00g_rr = -1` (metric `-b\,dt^2 + b^-1dr^2`), is `kappa = 1/2d_r b` at the horizon; on the rule's *vacuum* solution `b = 1 - r_s/r` (the linear rule is 3D-harmonic in vacuum) the horizon sits at `r = r_s` and `kappa = 1/(2r_s)`, so with `r_s ~ M` measured, **`kappa ~ 1/r_h`** — the Hawking relation. *(An earlier round reported this "flat" — a measurement error: it used `d_rsqrtb`, which **diverges** at the horizon since `sqrtb = N -> 0`; the corrected `kappa = 1/2d_r b` is finite and scales. The minimal rule carries the Hawking scaling; no hyperbolic strong-field rule is required.)* The honest residual is that a **fully-direct** simulation of the near-horizon slope is resolution-limited — the elliptic relaxation smooths the `b = 0` clip over a `D`-set width, capping the *measured* slope at the small horizons compact sources allow — so `kappa ~ 1/r_h` is tiered as **derived** (rule vacuum solution + GR-I relation + measured `r_s ~ M`), not as a clean direct measurement. The thermodynamic coefficient (`T = kappa/2pi`, `S = A/4`) stays value-inherited.

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9. Position Relative to the Quartet

- **GR-I's two assumptions are closed.** `alpha = 1` is forced (§4); the timelike geodesic identity is limit-forced (§5). GR-I's preamble items 6 and 9 are downgraded from postulate/assumption to derived/reduced, citing this paper.
- **GR-II's reduced object is built.** The field equation = steady state of `F` (R9) is now an `F` that exists and runs (§7); the mode count is reconfirmed from `F` (§7); the field-equation form's Newtonian limit is measured.
- **KM-II §6 is resolved.** The scalar-mode speed, left open, is `c_s = c` (§6, derived). KM-II's §6 caveat becomes "the khronon is at light speed, derived."
- **A2-dynamical (the working-program target #8) is realized.** The same rule that builds the EFE horizon forms A2's frozen cut — by build-and-run (§7).

What this paper does **not** change: the gravitational class (khronometric, GR-II); the MOND/cosmology sectors (KM-I/II); the value-inherited constants; and the one open number, `alpha_1, alpha_2` (§8).

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10. The Wedge — Empirical Content

1. **The khronon at light speed** — a falsifiable, sharp prediction: ED's scalar GW polarization propagates at `c` (not a fast/slow second cone), damped near matter, clean in vacuum. A scalar polarization at `c` is distinguishable in principle from pure-GR (two tensor modes only) and from generic khronometric gravity (`c_s != c`).
2. **The emergent horizon's area law** — `S ~ A` on a dynamically-formed horizon; the holographic scaling is a structural prediction, the `1/4` value-inherited.
3. **The preferred-frame numbers** — `alpha_1, alpha_2`, once computed (the definite, knob-free calculation §8 sets up), are the GR-II falsification front; this paper sharpens the lean and defines the computation, it does not pre-empt the verdict.
4. **Not claimed as wedges:** the `alpha_1, alpha_2` values; the Hawking `T` *coefficient* (the scaling is derived; the `T = kappa/2pi` normalization is value-inherited); the strong-field metric; the full EFE.

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11. Falsification Criteria

GR-III is falsifiable at every closed step and at each open number: break the lapse fork, the geodesic reduction, the khronon-speed derivation, or the field-equation/horizon build, and the corresponding result fails; the preferred-frame front closes for or against the theory once its numbers are computed. The sharpest, stated individually:

11.1 The lapse fork

Falsifier sentence: *A demonstration that the P04 commitment-reserve band co-scales with the internal band (`beta != 0`) — e.g. a replenishment channel slaving the reserve to `b_int` — would reopen `alpha != 1` and the Einstein/Nordström fork, falsifying §4.*

11.2 The geodesic reduction

Falsifier sentence: *An explicit Σ-rule computation showing the massive-front coherence functional is **not** the Jacobi `J(b)` (i.e. the massive worldlines are not timelike geodesics of `g ~ b^-1`) would falsify §5 — the limit checks (Fermat, Maupertuis) being necessary but not sufficient.*

11.3 The khronon speed

Falsifier sentence: *A demonstration that ED's foliation/reserve sector supplies a conservative `lambdatheta^2` term (`eps != 0`) — i.e. that the reserve coupling is kinetic, not dissipative — would falsify `c_s = c` (§6) and give the khronon a second cone.*

11.4 The field equation / horizon

Falsifier sentence: *A demonstration that the forced rule's steady state is **not** `grad ^2b ~ rho`, or that strong coupling does **not** form a finite-radius `b -> 0` horizon / the severed entropy scales as the bulk (`r_h^2`) not the surface (`r_h`), would falsify §7.*

11.5 The preferred-frame front (when computed)

Falsifier sentence: *Computed `alpha_1` or `alpha_2` outside the lunar-laser-ranging / pulsar bounds (~10^-4, ~10^-7), once **`kappa, D`** are pinned and the PPN expansion performed, would falsify ED-gravity-as-viable — the front this paper leaves open.*

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12. Position Statement

The dynamical rule Event-Density gravity reduces to is built and runs; its two weak-field assumptions are closed; its scalar mode is pinned to the light cone — and one irreversibility does all three structural jobs. `alpha = 1` is forced from the P04 band law (the reserve cannot replenish to track the metric band — Nordström excluded); the timelike geodesic identity is reduced to the massive eikonal of GR-I's proven null result; and the khronon propagates at `c` because the reserve sector is dissipative, not kinetic (KM-II §6 resolved). The rule, built, has the Newtonian field equation as its fixed point, the Schwarzschild relation `r_s ~ M` (2D and 3D), an emergent finite-radius horizon that is a frozen A2 cut carrying area-law entropy, and the khronometric mode count read off `F`. The same primitive — the arrow, P11 — selects the Einstein branch at the lapse, makes the khronon physical at the mode count, and keeps it on the light cone.

What this paper claims. The dynamical rule is built and Newtonian-consistent; `alpha = 1` is forced (Nordström excluded); the timelike geodesic identity is limit-forced; `c_s = c` is derived; the emergent horizon, frozen A2 cut, and area law are measured; the mode count is from `F`; and the arrow's three structural selections are one unification.

What this paper does not claim. The full EFE is not derived; the class is khronometric, not pure GR; `alpha = 1` rests on a band-accounting premise; the geodesic identity is reduced, not proven; `c_s = c` is leading-order; the rule's coefficients are unpinned; **`alpha_1, alpha_2` are not computed (and not faked)**; the Hawking scaling is **derived** (`kappa = 1/(2r_s) ~ 1/r_h`) but its **direct** simulation is resolution-limited and its coefficient value-inherited; and Einstein is not derived.

Series context. Paper GR-III of the post-pivot curvature-emergence line — the engine of the quartet (GR-I, GR-II, KM-I, KM-II): the rule they reduce to, built; the assumptions they carried, closed; the scalar speed they left open, derived. Two falsification numbers remain open, stated as such.

Appendix: Cross-References, Glossary, Reader Map

Cross-references

- **Paper_087:** position paper. — **Paper_027:** Newton's `G` (value-inherited).
- **GR-I:** weak-field metric (assumptions closed here). — **GR-II:** khronometric class; `F`-reduction (built here).
- **KM-I/KM-II:** deep-field + cosmology (KM-II §6 resolved here).
- **Working record:** `event-density/foundations/` (the Phase-3 GR rounds and the dynamical-rule builds) and `evaluation/DynamicalBandwidth/`.

Glossary

- **The dynamical rule `F`.** `b-dot = F(rho,grad rho,b)` — the bandwidth evolution whose steady state is the field equation (GR-II/R9); minimal forced form `b-dot = Dgrad ^2b - kapparho`.
- **The three hats (of the arrow).** P11 selecting the Einstein/khronometric structure three times: `alpha=1` (lapse), khronon physical (mode count), khronon at `c` (scalar speed).
- **`alpha = 1` / P-Commitment-Linear.** The lapse exponent, forced by the reserve's inability to replenish (P11); Nordström excluded.
- **Jacobi functional.** `J(b) = b^-1sqrt(E^2-m^2b)` — the timelike-geodesic functional; limit-forced to Fermat (`m->0`) and Maupertuis (weak field).
- **`eps / c_s = c`.** The foliation-kinetic knob (`c_s/c = sqrt(1+eps)`); derived zero (dissipative reserve), so the khronon is at light speed.
- **Frozen A2 cut.** The emergent `b->0` horizon: a metric horizon, an A2 decoupling cut, and a V5 surface — one locus.
- **Area law `S ~ A`.** The severed-channel count (A1) scaling as the horizon perimeter, not the bulk — holographic; coefficient value-inherited.

Reader map and open work

Where to look next. - *The weak-field metric and tests:* GR-I. — *The class and GW/Lorentz:* GR-II. — *MOND/cosmology:* KM-I/KM-II. - *The build details:* `event-density/foundations/Phase3_GR_*` and `evaluation/DynamicalBandwidth/`.

Open work (declared): **compute `alpha_1, alpha_2`** (pin `kappa,D` $\to$ khronometric PPN, or direct PPN expansion of `F`); a **fully-direct** simulation of the Hawking surface gravity (the scaling is derived; the direct measurement is resolution-limited — needs large horizons or a non-relaxational strong-field rule); the explicit Σ-rule massive-front computation (closes §5); the sub-leading reserve speed-renormalization; pinning `kappa, D`.

End of Paper GR-III.

Post-pivot curvature-emergence line, the engine of the quartet. The dynamical rule ED's gravity reduces to is built and runs (Newtonian field equation as its fixed point; Schwarzschild `r_s~M` in 2D and 3D; an emergent finite-radius horizon that is a frozen A2 cut carrying area-law entropy `S~A` and the Hawking scaling `kappa=1/(2r_s)~1/r_h`; the khronometric mode count read off `F`). Its two weak-field assumptions are closed: `alpha=1` forced from the P04 band law (Nordström excluded), the timelike geodesic identity limit-forced to GR-I's proven null result. Its scalar mode is pinned to the light cone (`c_s=c`, derived — the reserve is dissipative, not kinetic; KM-II §6 resolved). One irreversibility (P11) does all three structural jobs — the lapse, the mode count, the scalar speed. One number stays open and is flagged: `alpha_1,alpha_2` (structurally suppressed, not computed, not faked); the Hawking scaling, earlier thought open, is corrected to a derived result (the direct sim measurement remains resolution-limited). No new primitives; Einstein not derived; the engine exists.